

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

Paper 2 Core Paper

9648/02

September 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

This document consists of **18** printed pages and **no** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 The electron micrograph in **Fig. 1.1** shows an acinar cell found in the pancreas.

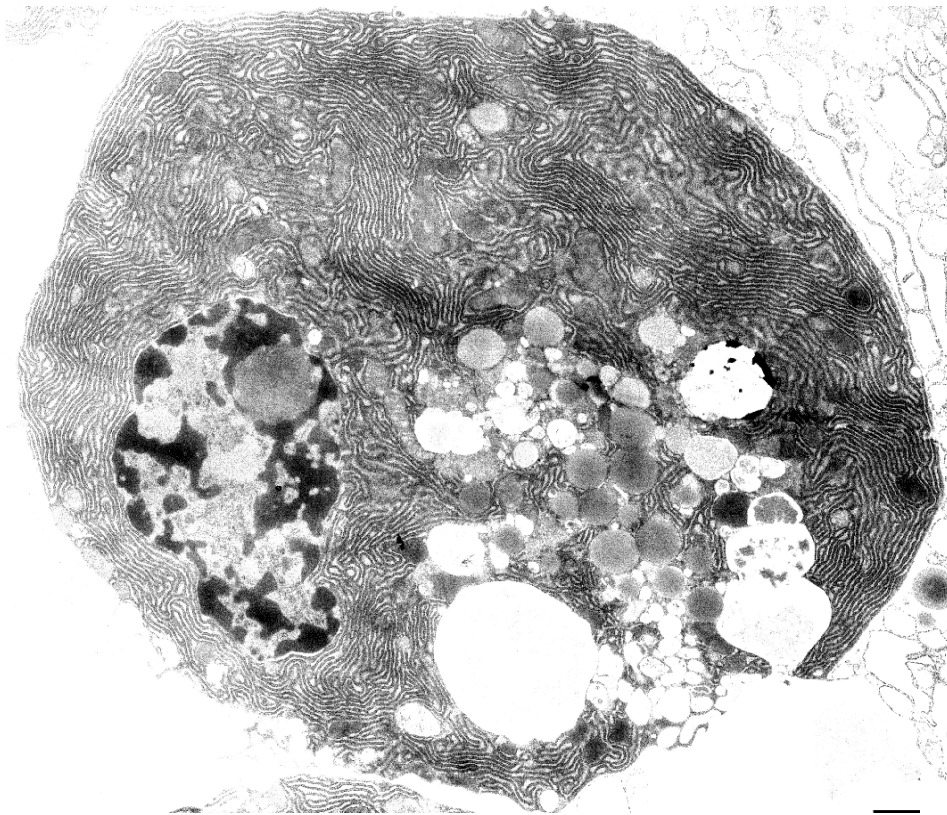


Fig. 1.1

- (a) With reference to the **Fig. 1.1**, suggest the principal function of the pancreatic acinar cell. Give **two** reasons to explain your answer.

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[3]

Fig. 1.2 shows two different structures, **A** and **B**, found in an animal cell.

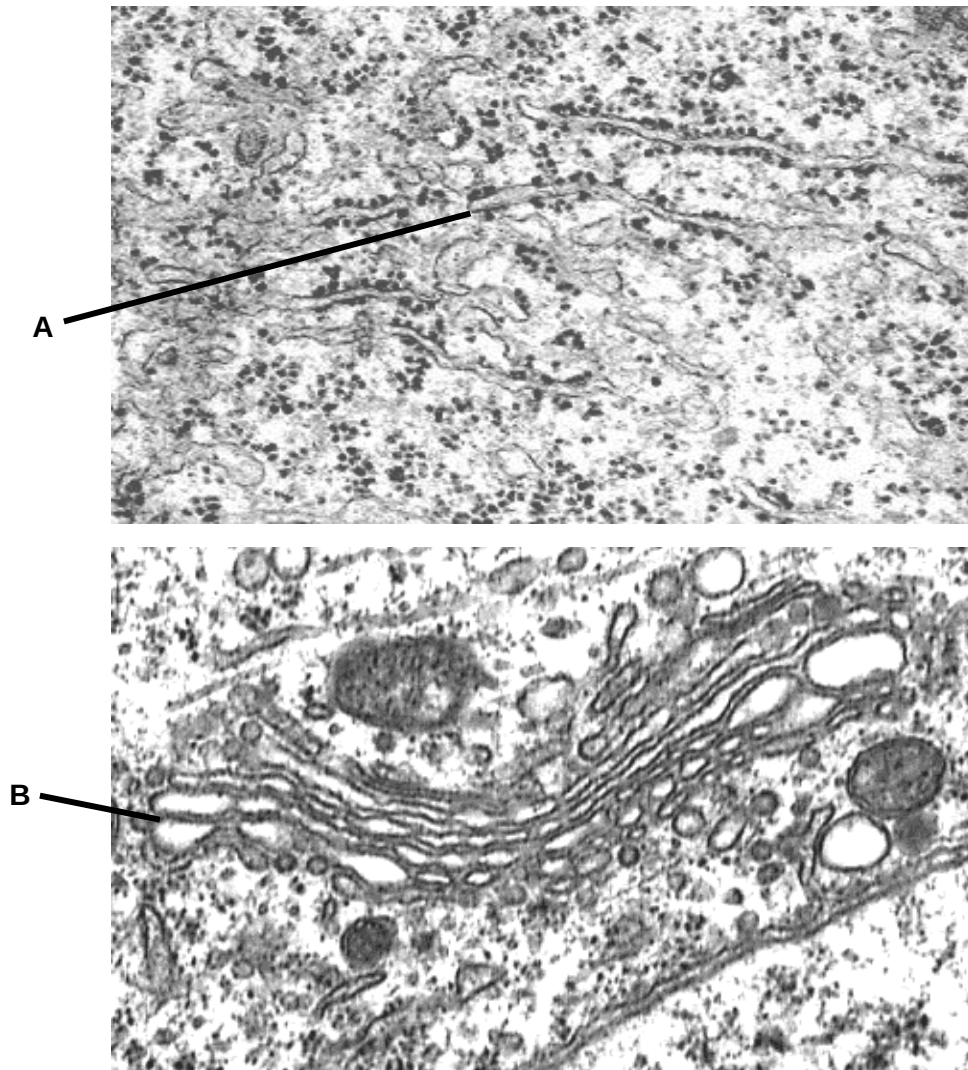


Fig. 1.2

(b) With reference to the **Fig. 1.2**, describe **three** ways in which **A** is similar to **B**.

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[3]

Scientists isolated mitochondria from the liver cells. They broke the cells open in an ice-cold, isotonic solution. They then used a centrifuge to separate the cell organelles. **Fig. 1.3** illustrates the steps that take place in the process of centrifugation.

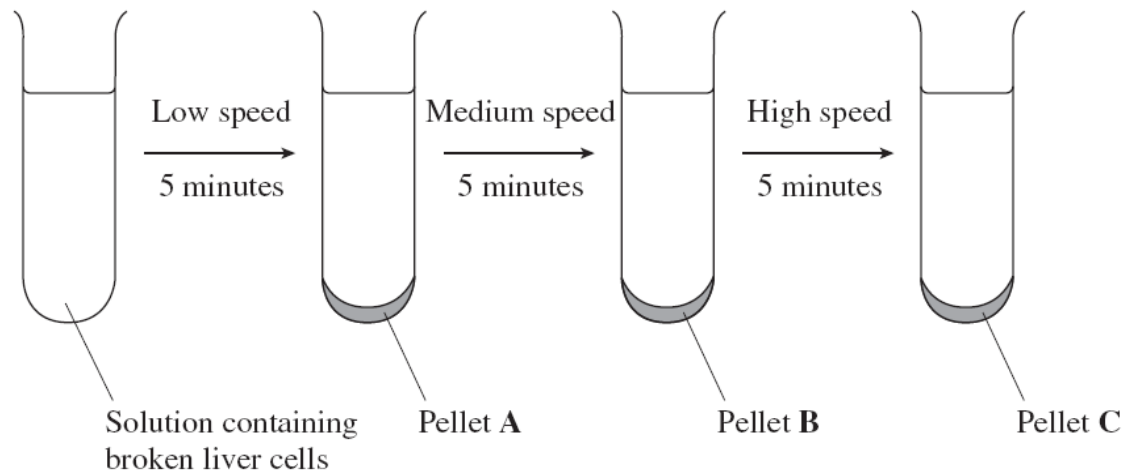


Fig. 1.3

(c) State and explain which pellet contains mitochondria.

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[2]

(d) Suggest why the solution used was ice-cold and isotonic.

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[2]

People with mitochondrial disease have mitochondria that do not function properly. These disorders may be caused by mutations in the mitochondrial DNA or in the nuclear genes that code for mitochondrial components.

(e) Explain why a person with mitochondrial disease can only exercise for a short time.

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[2]

[Total: 12]

- 2 Prokaryotes were first discovered in the late 1600s by Antony van Leeuwenhoek, using the microscope he invented. With the advancements in science, scientists have been able to unravel the mysteries between prokaryotes and eukaryotes.

Fig. 2.1 (a) and (b) shows protein synthesis carried out by the two types of cells.

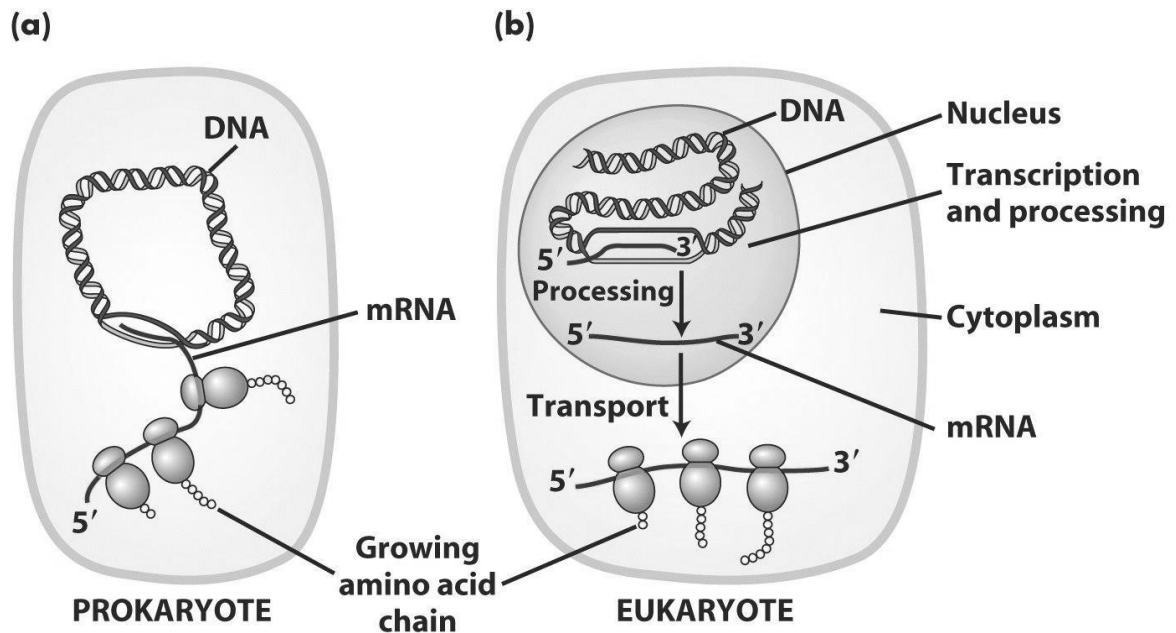


Fig. 2.1

- (a)** Describe the structure of a RNA molecule that is not shown in **Fig. 2.1** in relation to its function.

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[3]

- (b)** Describe the differences between prokaryotes and eukaryotes with respect to gene expression at the following levels:

- (i)** Chromatin level

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[2]

- (ii)** Transcriptional level

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[3]

During research into the mechanism of DNA replication, bacteria were grown for many generations in a medium containing only the 'heavy' isotope of nitrogen, ^{15}N . This resulted in all the DNA molecules containing only ^{15}N . This is illustrated in **Fig. 2.2** below.

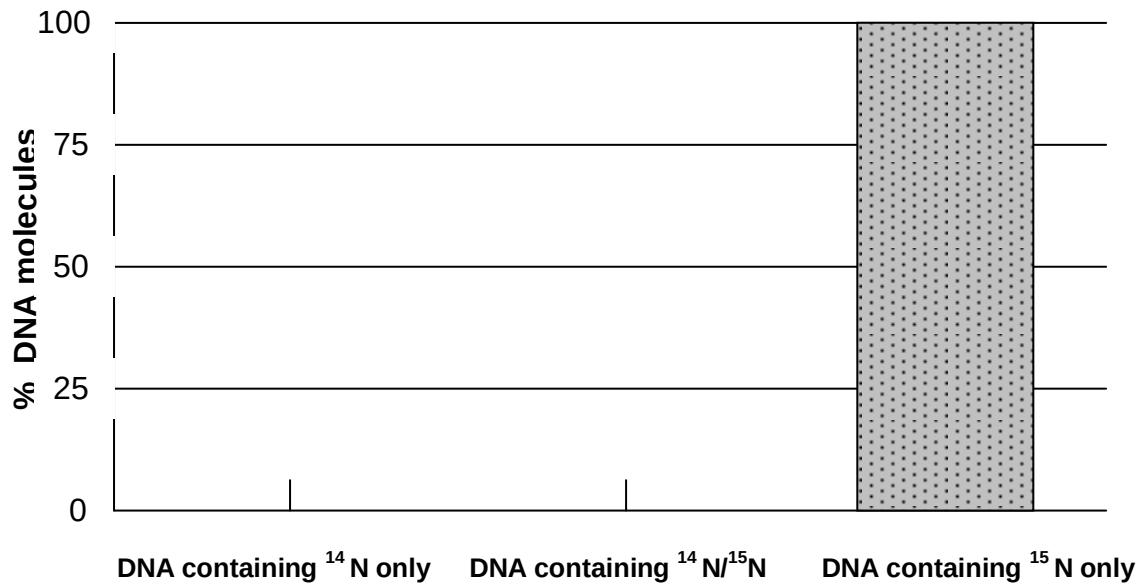


Fig. 2.2

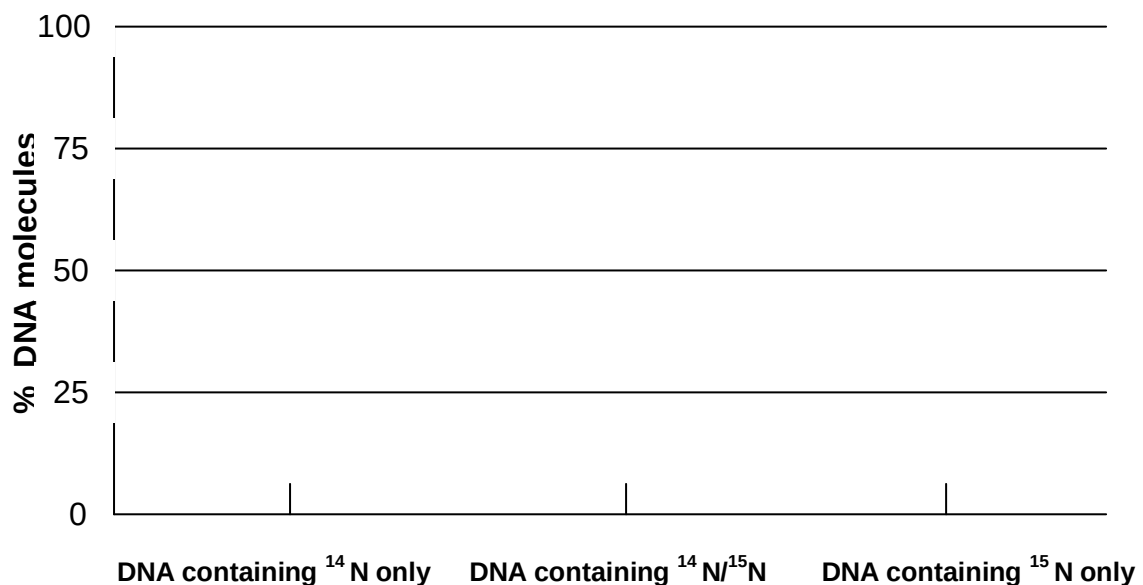
Step 1: These bacteria were then grown in a medium containing only 'light' nitrogen, ^{14}N . After the time taken for the DNA to replicate once, the DNA was analysed.

Step 2: The bacteria continued to grow in the 'light' nitrogen, ^{14}N , medium until the DNA had replicated once more. The DNA molecules were then analysed.

Step 3: The bacteria was allowed to grow in the 'light' nitrogen, ^{14}N , medium for another round of DNA replication.

(c) Complete the bar chart below to indicate the expected results of the composition of these DNA molecules after **step 3**.

[1]



DNA replication is a process which involves DNA polymerase and the deoxyribonucleotides A, T, C and G, while transcription involves RNA polymerase and the ribonucleotides A, U, C and G.

(d) Describe two **other** differences and two similarities between DNA replication and transcription.

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[4]

[Total: 13]

- 3 **Fig. 3.1** is an electron micrograph showing T4 bacteriophages invading a bacterium.

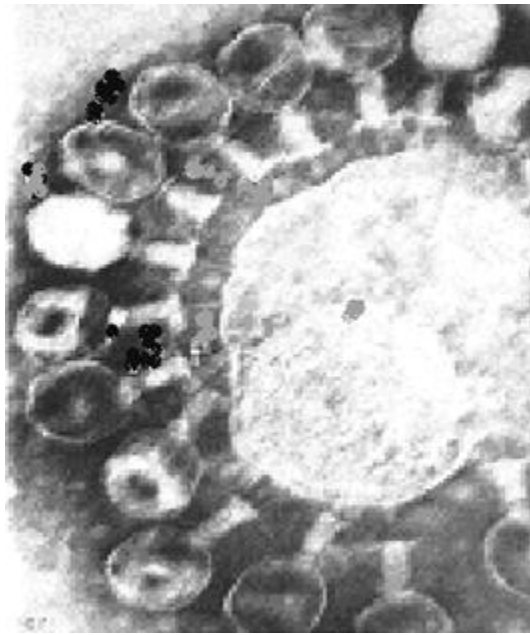


Fig. 3.1

- (a) Clearly label the base plate of one T4 phage in **Fig. 3.1**. [1]
- (b) Explain how T4 may mediate the transfer of a random piece of this bacteria's genetic material to another bacteria.

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[4]

- (c) Briefly describe the structure of bacterial chromosomes.

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[2]

Some genes in bacteria are organized into operons, such as the *lac* operon. Transcription of the *lac* operon in *E. coli* is dependent on the presence or absence of food molecules such as glucose and lactose.

- (d) **Under normal conditions**, predict the transcription of the operon in the presence or absence of these two substrate molecules by writing “**High**” or “**Low**” in the table below.

Lactose	Glucose	Transcription
+	+	
+	-	
-	+	
-	-	

[1]

- (e) **Mutation XX-6** deletes the DNA binding site from the CAP protein, but leaves the rest of the operon unaffected. Predict the effect on transcription under these four conditions by writing “**High**” or “**Low**” in the table below.

Lactose	Glucose	Transcription
+	+	
+	-	
-	+	
-	-	

[1]

- (f) Another mutation on a separate site gives the following results.

Lactose	Glucose	Transcription
+	+	Low
+	-	High
-	+	Low
-	-	High

Suggest the mutation that could have occurred and explain how it would have given rise to the above results.

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[2]

[Total: 11]

- 4 A woman's risk of developing breast cancer is greatly increased if she inherits *BRCA1* gene mutation. The role of *BRCA1* protein is shown in **Fig. 4.1**.

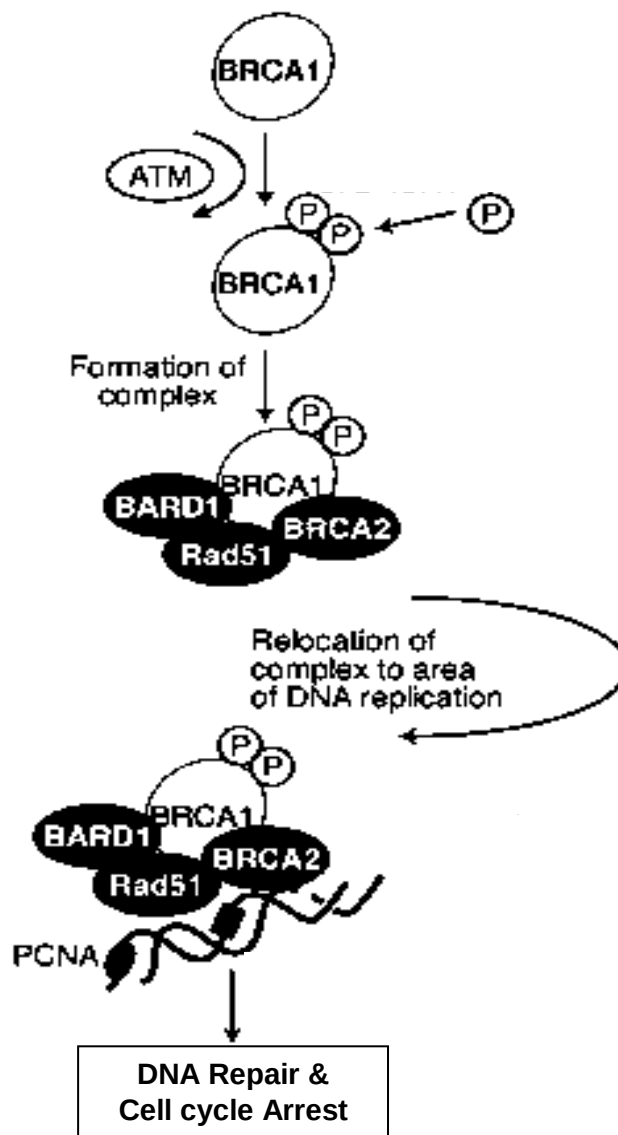


Fig. 4.1

- (a) With reference to **Fig. 4.1**,
 (i) State the class of genes that *BRCA1* belongs to.

[1]

(ii) Explain how mutations in *BRCA1* gene can contribute to cancer.

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[3]

(iii) The expression of *BRCA1* gene can be furthered controlled at post-translational level. Explain how this can occur.

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[2]

(b) The *HER2* gene codes for a growth factor receptor. When the growth factor binds to HER2 protein (the growth factor receptor), the signal is given for the cell to start dividing. Gene amplification of *HER2* gene could also result in breast cancer.

(i) Describe a process which results in gene amplification.

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[2]

(ii) Suggest how amplification of *HER2* could result in breast cancer.

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[2]

[Total: 10]

- 5 Rickets is a childhood disorder involving the softening and weakening of bones in humans. It is usually caused by a lack of vitamin D, calcium ions or phosphate ions. A rare form of rickets that cannot be successfully treated with vitamin D therapy is caused by an allele on the X chromosome that arose as a result of a gene mutation.

(a) Explain what is meant by the term gene mutation.

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[2]

(b) Fig. 5.1 below illustrates how this rare form of rickets is inherited in an afflicted family.

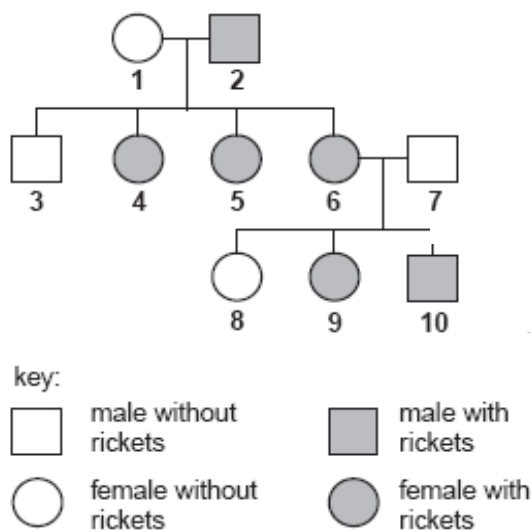


Fig. 5.1

- (i) With reference to **Fig. 5.1**, state and explain the pattern of inheritance of this form of rickets.

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[3]

- (ii) Using appropriate symbols, state the genotypes of the following individuals.

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3

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9

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10

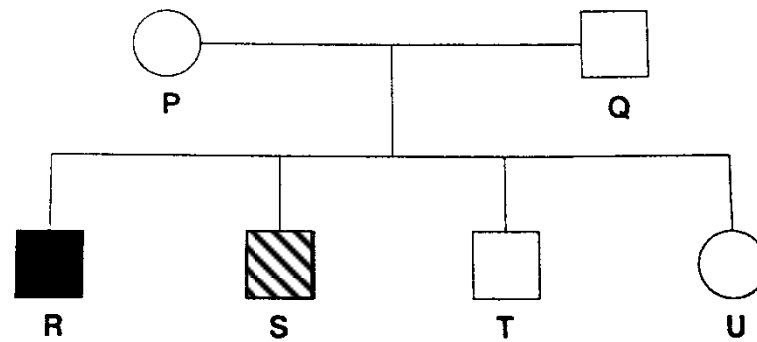
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[2]

- (iii) Couple 6 and 7 decided to have another son. Draw a genetic diagram to determine the probability that their second son will suffer from rickets.

[3]

Colour blindness and haemophilia are inherited sex-linked recessive conditions in humans. **Fig. 5.2** shows two generations of a family in which these conditions are inherited.



Key:



Fig. 5.2

Suggest how the genotype of **S** has arisen.

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[2]

[Total: 12]

- 6 **Fig. 6.1** below shows the cell signalling pathway in smooth muscle cells in response to the hormone Epinephrine, resulting in smooth muscle relaxation of blood vessels (vasodilation).

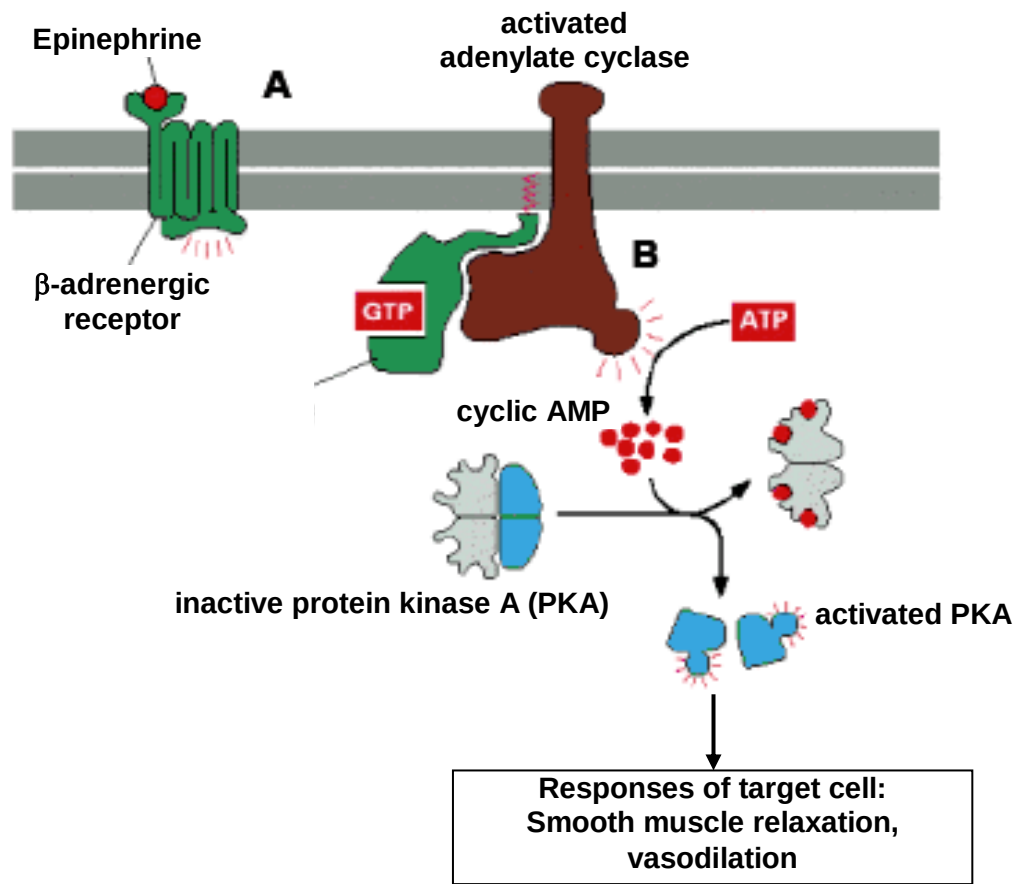


Fig. 6.1

- (a) State the class of receptors that the β -adrenergic receptor belongs to.

[1]

- (b) With reference to **Fig. 6.1**,
(i) describe how adenylate cyclase is activated.

[3]

- (ii) identify the stage where signal amplification is observed and explain how the signal is amplified in that stage.

[2]

- (c) Explain how the structure of the β -adrenergic receptor allows it to carry out its function as a receptor.

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[3]

- (d) A drug was designed to ensure that permanent vasodilation occurs. Suggest how this drug might work within the cell signalling pathway in **Fig. 6.1**.

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[2]

[Total: 11]

- 7 Scientists studying the evolutionary history of the Human Immunodeficiency Virus (HIV) found that it is closely related to the Simian Immunodeficiency Virus (SIV) found in certain species of monkeys. The phylogenetic tree in **Fig. 7.1** shows part of the evolutionary relationships between these viruses. A common ancestral virus is believed to have evolved into strains that infected chimpanzees (SIV). Over time, new strains emerged which infect humans.

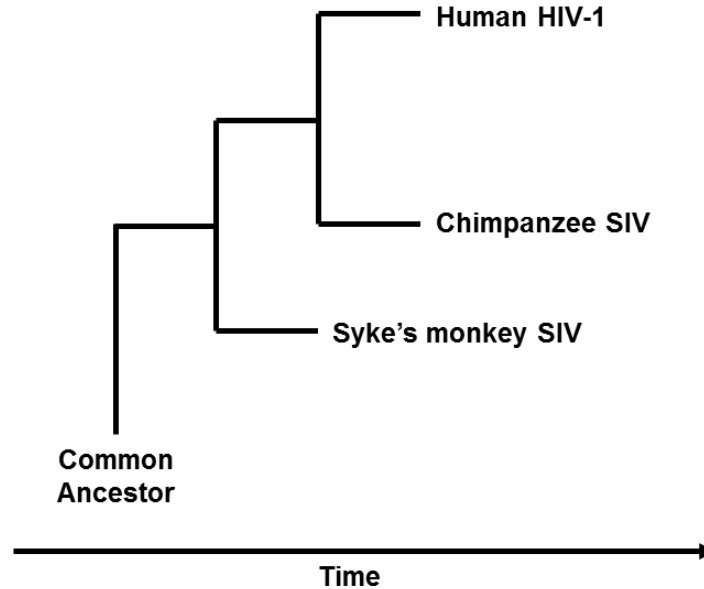


Fig. 7.1

- (a) Explain the significance of the branch points in the phylogenetic tree shown in **Fig. 7.1**.

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[2]

- (b) Explain how molecular homology could be used to support the claim that HIV and SIV share an evolutionary relationship.

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[2]

(c) State **two** advantages of using molecular techniques in classification.

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[2]

SIV infections in their natural hosts appear in many cases to be non-pathogenic (non-disease causing). Certain individuals within the human population have also shown similar characteristics when exposed to HIV. These individuals continue to test negative for HIV despite being repeatedly exposed to HIV.

Investigations have found a mutation present in the CCR5 gene in these individuals. CCR5 is a co-receptor that facilitates the entry of HIV into the T-cell after gp120 (present on the viral envelope) binds with the CD4 receptor.

(d) Suggest and explain how these individuals continue to resist HIV infection.

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[2]

(e) Assuming that humans do not take active steps to prevent HIV infection, explain how the human population could possibly evolve to not be affected by HIV.

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[3]

[Total: 11]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8** **(a)** Outline how the endomembrane system is involved in synaptic transmission. [8]
 (b) Describe the main stages of the Calvin cycle [8]
 (c) Contrast the structure of cellulose with that of collagen and suggest why plant cells have cellulose but not collagen. [4]
- 9** **(a)** Compare and contrast the structures of influenza and HIV. [8]
 (b) Describe how HIV infects host cells and reproduces. [8]
 (c) Briefly explain how a person infected only **once** by one strain of HIV may eventually have many strains of HIV. [4]